

PHYS 105 – Practical Physics: Mechanics and Thermal Physics

Uncertainty analysis

How good is this number, and how do I show it?

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Reading: OpenStax *University Physics* Vol. 1, §1.6; Taylor, *An Introduction to Error Analysis*, Ch. 1–4 and 8; Hughes and Hase, *Measurements and Their Uncertainties*, Ch. 1–6

By the end of this lecture, you should be able to:

1. Report a result as (value $\pm$ uncertainty) unit, correctly rounded, with its percent uncertainty.
2. Distinguish accuracy, precision, random and systematic uncertainty, and classify a source as Type A or Type B.
3. Calculate the mean, standard deviation and standard error of repeated readings.
4. Propagate uncertainties through a formula and find the dominant source.
5. Fit a straight line, interpret its residuals and χ^2 , and report slope and intercept.
6. Build an uncertainty budget and test agreement with an accepted value.

Three contacts:

1 UNDERSTAND

What is a result, and where does uncertainty come from?

2 QUANTIFY

How do repeated readings and formulas give a number with an uncertainty?

3 APPLY

Graphs, budgets, comparison with theory, and the report.

Not marked. Answer individually, then we review.

1. How many significant figures are in 0.0450, in 4.50×10^3 and in 1020?
2. What percentage is 0.5 of 40?
3. Round 7.846 to two decimal places and to two significant figures.
4. A ruler is marked in millimetres. What is the smallest length you can read directly?
5. The slope of a straight-line graph of y against x has what units?
6. Differentiate $f = x^2$ and $f = \sin x$.

1. 3, 3, 4. Leading zeros are placeholders; the trailing zero in 0.0450 is significant.
2. $0.5/40 = 1.25\%$.
3. 7.85; 7.8.
4. 1 mm, with the last digit estimated by eye.
5. Units of y divided by units of x .
6. $2x$; $\cos x$ (with x in radians).

Contact 1: Understand

*What is a measurement result, and
where does uncertainty come from?*

Two students measure the length of the same bench

Student X reports: “the bench is 1.2 m long.”

Student Y reports: “the bench is 1.213 ± 0.002 m long.”

Predict first

Which report is more useful to a third person who must cut a shelf to fit? What does Y tell you that X does not? Is either report certainly *true*?

- X gives one number and no sense of how far to trust it.
- Y gives a best value and a range. The shelf can be cut to about 1.21 m with a known margin.
- Neither report is certain: every measurement has an uncertainty, and **saying so is part of the result.**

A measurement is not complete without its uncertainty

Rule: the form of a measurement result

A result has three parts: a **best estimate**, an **uncertainty** and a **unit**:

$$x = (x_{\text{best}} \pm \delta x) \text{ unit}$$

δx is the **uncertainty**: the size of the range, around the best estimate, within which the true value is expected to lie with reasonable confidence.

Physical meaning

“Uncertainty” is not “mistake”. A mistake (reading the wrong scale, writing the wrong digit) can be avoided. Uncertainty cannot be avoided; it can only be *estimated* and *reduced*. It is the honest statement of what the measurement can and cannot tell us.

Notation follows OpenStax §1.6 ($A \pm \delta A$). δx is one *standard uncertainty* (later slides).

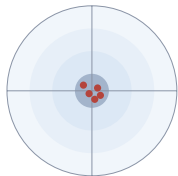
Vocabulary: four words that are often confused

Term	Meaning	Needs a reference value?
Accuracy	How close a measurement is to the accepted reference value.	Yes
Precision	How closely repeated measurements, under the same conditions, agree with each other.	No
Uncertainty δx	A quantitative measure of the spread or doubt in the result. Related to precision.	No
Discrepancy	The difference between the measured value and the accepted value: $x_{\text{best}} - x_{\text{acc}}$. Related to accuracy.	Yes

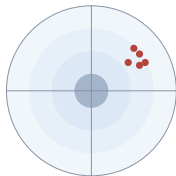
Caution

In everyday speech “error” means a mistake. In practical physics the *difference from the accepted value* is the discrepancy, and the *doubt in the result* is the uncertainty. Neither is a mistake.

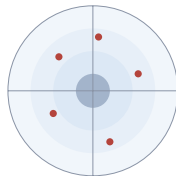
Sketch first: accuracy and precision



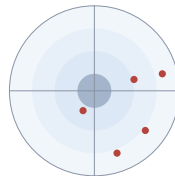
(a) **precise, accurate**
small spread, centred



(b) **precise, not accurate**
small spread, offset



(c) **accurate, not precise**
large spread, centred



(d) **neither**
large spread, offset

Predict first

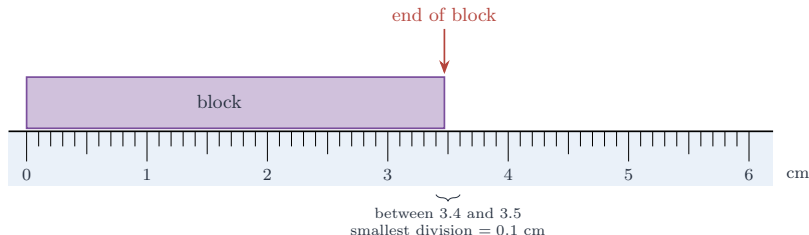
Each dot is one measurement of the same quantity; the centre is the accepted value. Before reading the labels: which panel has small uncertainty and small discrepancy? Which has small uncertainty but large discrepancy? What kind of problem would produce panel (b)?

- Panel (a): the ideal. Both small.
- Panel (b): the scatter is small but everything is shifted. A *consistent* cause, such as a worn ruler end, a clock that runs fast, or a thermometer that reads 1°C high. Repeating more times will **not** remove it.
- Panel (c): scatter large but centred. Averaging helps (Contact 2).
- Panel (d): a scatter problem and a shift.

Why it matters

Precision can be estimated from the data alone. Accuracy cannot: it needs an independent reference value or a different method. So **high precision does not guarantee accuracy**.

Reading an instrument



Which value should be written down?

The end of the block lies between 3.4 and 3.5 cm. We read 3.47 cm by estimating where it falls between two marks. The last digit is an *estimate*. How uncertain is it?

TOOLKIT 1 – Reading an instrument

Resolution is the smallest change the instrument can show. It sets a lower limit on the uncertainty of one reading.

1. **Analogue scale:** read to a fraction of the smallest division. A safe default is a reading limit of **half the smallest division**. State a smaller value only if you can justify it.
2. **Digital display:** the limit is at least **one unit in the last digit**, unless the manufacturer states otherwise.
3. **Write** the unit and every digit the instrument gives, including a final zero.
4. **Zero:** check the reading with nothing being measured (jaws closed, balance empty).

WATCH OUT: the reading limit is only one source. It is not the whole uncertainty of the measurement.

Concept check: how uncertain is each reading?

Assign a reading limit and write each result in full

1. Metre rule, smallest division 1 mm, block end at 3.47 cm.
2. Digital balance displays 152.4 g.
3. Stopwatch displays 15.82 s (resolution 0.01 s).
4. Analogue thermometer, 1 °C divisions, mercury level halfway between 23 and 24 °C.

Predict first

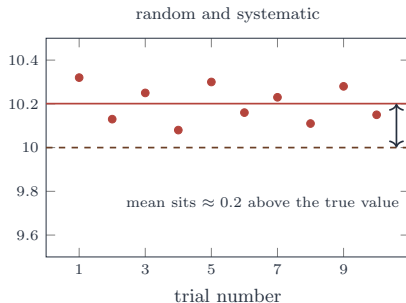
Before looking at the next slide: in which of the four is the instrument's resolution *not* the main concern in practice?

1. Half of 1 mm is 0.5 mm = 0.05 cm: (3.47 ± 0.05) cm. The second decimal is already uncertain, so do not quote 3.470.
2. Last digit is 0.1 g: (152.4 ± 0.1) g.
3. Last digit is 0.01 s: (15.82 ± 0.01) s from the display. But a hand-started stopwatch is limited by *reaction time*, typically 0.1 to 0.2 s, far larger than the display. This is the answer to the prediction.
4. Half of 1 °C is 0.5 °C: (23.5 ± 0.5) °C.

Lesson

The resolution is a *minimum* uncertainty. The real uncertainty is set by the largest source present.

Sketch first: do repeated readings remove the uncertainty?



Predict first

Ten readings of the same quantity are shown in two situations. If we average each set of ten, where does each average land relative to the true value? If we take 1000 readings instead of ten, what changes in each panel, and what does not?

Rule: two kinds of uncertainty

Random: unpredictable, equally likely to be above or below the true value, changes from reading to reading. *Averaging reduces it.*

Systematic: shifts every reading the same way (same size and direction). *Averaging does not reduce it.* It is found by calibration, by comparison with another method, or by checking the instrument, and is then corrected.

Typical random sources

Reaction time when starting and stopping

Judging a pointer or the end of an object

Electrical noise, air draughts

Small variations in the object itself

Typical systematic sources

Zero error (micrometer closed reads +0.03 mm)

Instrument out of calibration

Parallax from always viewing from one side

A false assumption in the method

Rule: how an uncertainty is evaluated (the GUM classification)

Type A: evaluated from the *statistics of repeated measurements* (the scatter of the readings; Contact 2).

Type B: evaluated by *any other means*: instrument resolution, a manufacturer's specification, a calibration certificate, a published value, or reasoned judgement.

Do not confuse the two classifications

Random against systematic describes the *physical behaviour* of the source. Type A against Type B describes the *method of evaluation*. A random source is usually Type A. A systematic source can only be Type B, because repeating does not reveal it. A resolution limit is Type B but behaves randomly.

Source: JCGM 100:2008 (GUM); NIST TN 1297. First-year laboratories often use only “random” and “systematic”.

Turning a reading limit into a standard uncertainty

Rule: from a limit $\pm a$ to a standard uncertainty

If a quantity is known only to lie within $\pm a$ of the reading, with no value in that range more likely than another (a *rectangular* distribution), the standard uncertainty is

$$u = \frac{a}{\sqrt{3}} \approx 0.58 a.$$

Instrument	Limit a	$u = a/\sqrt{3}$
Metre rule, 1 mm divisions	0.5 mm	0.29 mm
Digital stopwatch, 0.01 s	0.01 s (one unit in the last digit)	0.0058 s
Micrometer, 0.01 mm	0.005 mm	0.0029 mm

Check what your laboratory manual asks

Some laboratories quote the limit a itself as the reading uncertainty. That is acceptable if applied consistently, but a limit must not be combined with a standard deviation as if both were standard uncertainties. This lecture uses $u = a/\sqrt{3}$ when they are combined.

✂ TOOLKIT 2 – Classifying a source

For each source of uncertainty, ask in order:

1. **Would repeating change it?** Random (varies) or systematic (same each time).
2. **Can I estimate it from repeated readings?** Type A (use scatter) or Type B (use limit, specification, judgement).
3. **How large is it?** Scatter: $s/\sqrt{N}$. Limit $\pm a$: $a/\sqrt{3}$.
4. **Can I correct it?** If a systematic shift is known, *correct* the value and keep a (small) uncertainty on the correction.

Pendulum source	Random/systematic	Type	Estimate
Scatter of $20T$ timings	random	A	$s/\sqrt{N}$
Stopwatch resolution	random-like	B	$0.01/\sqrt{3}$ s
Length L (locating bob centre)	either	B	judgement, 0.002 m
Finite swing amplitude	systematic	B	correct, then $\pm$

Worked example: zero error and calibration

IDENTIFY. A micrometer reads $+0.03\text{ mm}$ with its jaws closed (zero error). It reads 5.47 mm on a rod. A thermometer reads 1.5°C in melting ice and 98.5°C in steam at 100.0°C ; it reads 36.8°C on a sample. Required: the corrected values.

SET UP. Zero error: a *constant offset*, so subtract it. Thermometer: two known points fix a straight line from reading to true value.

Prediction: the rod is slightly thinner than 5.47 mm ; the thermometer reads *high* at the low end and *low* at the high end, so the sample is below 36.8°C .

EXECUTE.

Rod: $5.47 - 0.03 = 5.44\text{ mm}$.

$$\begin{aligned}\text{Thermometer: } T_{\text{true}} &= \frac{T_{\text{read}} - 1.5}{98.5 - 1.5} \times 100.0 \\ &= \frac{36.8 - 1.5}{97.0} \times 100.0 = 36.4^\circ\text{C}.\end{aligned}$$

EVALUATE. The corrections (-0.03 mm ; -0.4°C) agree with the predictions. The thermometer correction is about as large as its 0.5°C reading limit, so it *cannot be ignored*. The corrections themselves carry uncertainty, which enters in Contact 2.

Can you now:

- say why “1.2 m” alone is an incomplete result?
- distinguish accuracy from precision on a target sketch?
- read an instrument and state its reading limit?
- say which uncertainties averaging removes, and which it does not?
- classify a source as random or systematic, Type A or Type B?

Next

How do we turn ten repeated readings into one number and its uncertainty, and how does an uncertainty in L and T become an uncertainty in g ? That is Contact 2.

Contact 2: Quantify

How do repeated readings and formulas give a number with an uncertainty?

An opening question: ten timings

A pendulum of length $L = 0.800$ m: ten timings of 20 oscillations, t_{20} in seconds

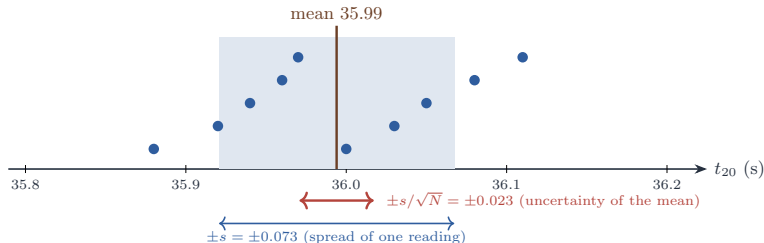
35.92 36.05 35.88 36.11 35.97 36.03 35.94 36.08 36.00 35.96

Predict first

Sketch the ten values on a number line. (i) Estimate the best value of t_{20} . (ii) Roughly how far does a typical reading lie from it? (iii) If a colleague repeats the whole set of ten, how far would their *mean* differ from yours: about the same as the scatter of one reading, or much less?

Timing 20 oscillations rather than one is itself a design choice: the stopwatch limit is shared among 20 periods. The data are illustrative.

Sketch: the data and what we summarise



Rule: summarising N repeated readings $x_1, \dots, x_N$

$$\bar{x} = \frac{1}{N} \sum_i x_i, \quad s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}, \quad \delta\bar{x} = \frac{s}{\sqrt{N}} \quad (\text{standard error of the mean}).$$

s is the spread of a *single* reading. $s/\sqrt{N}$ is the uncertainty of the *mean*.

TOOLKIT 3 – Summarising repeated data

1. **List and sketch** the readings. Look for a pattern (drift) and for a value far from the rest.
2. **Mean** $\bar{x}$: the best estimate.
3. **Standard deviation** s with $N - 1$ in the denominator (one degree of freedom is used up by $\bar{x}$).
4. **Standard error** $s/\sqrt{N}$: the Type A uncertainty of the mean.
5. **Report** $\bar{x} \pm s/\sqrt{N}$ (Toolkit rounding), and state N .

CHECK: about two thirds of the readings lie within $\bar{x} \pm s$.

WATCH OUT: a drift (readings steadily rising) means the conditions changed. Averaging a drift is meaningless.

Worked example: the pendulum timings

IDENTIFY. $N = 10$ readings of t_{20} . Required: best value and its uncertainty, then $T = t_{20}/20$.

SET UP. Toolkit 3. *Prediction:* the mean is near 36.0 s; the readings spread by about 0.07 s.

EXECUTE.

$$\bar{t} = 359.94/10 = 35.994 \text{ s},$$

$$s = \sqrt{0.0484/9} = 0.073 \text{ s},$$

$$\frac{s}{\sqrt{N}} = \frac{0.0734}{\sqrt{10}} = 0.023 \text{ s}.$$

Resolution (Type B): $0.01/\sqrt{3} = 0.0058 \text{ s}$.
Combine the Type A and Type B terms (rule on a later slide):

$$\delta t_{20} = \sqrt{0.0232^2 + 0.0058^2} = 0.0239 \text{ s}.$$

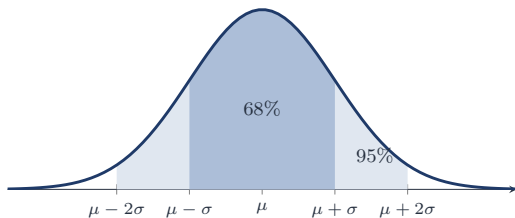
Divide by 20 (exact):

$$T = \frac{35.994}{20} = 1.7997 \text{ s},$$

$$\delta T = \frac{0.0239}{20} = 0.0012 \text{ s}.$$

EVALUATE. $\bar{t}$ and s match the predictions. The resolution term is 4 times smaller than the scatter and raises the total only from 0.0232 to 0.0239. $T = (1.7997 \pm 0.0012) \text{ s}$ (0.07%).

Why $s/\sqrt{N}$?



For scatter of the common bell-shaped (normal) kind, about 68% of single readings lie within $\bar{x} \pm s$ and about 95% within $\bar{x} \pm 2s$.

Rule: the $\sqrt{N}$ law

The standard error falls as $1/\sqrt{N}$, while s itself does *not* shrink as N grows.

N	$s/\sqrt{N}$ compared with $N = 1$
4	half
16	one quarter
100	one tenth

Predict first

The standard error in the pendulum example is 0.023 s. How many readings give 0.0115 s? ($4 \times 10 = 40$.) Is it worth it?

Six measurements of g (m/s^2): 9.78, 9.81, 9.75, 9.80, 9.77, 9.52

Is 9.52 a legitimate reading, or should it be discarded?

Rule: Chauvenet's criterion

Compute $\bar{x}$ and s from all N readings. For a suspect reading x_{sus} let $z = |x_{\text{sus}} - \bar{x}|/s$. The expected number of readings that far out is $N \cdot P(|Z| > z)$. If this number is **less than 0.5**, the reading may be rejected, and $\bar{x}$, s are recomputed without it.

Here $\bar{x} = 9.738$, $s = 0.109$, $z = 2.00$, so $N P = 6 \times 0.0455 = 0.27 < 0.5$: **reject**. Without it: $\bar{x} = 9.782$, $s = 0.024$, $s/\sqrt{5} = 0.011$, so $g = (9.78 \pm 0.01) \text{ m/s}^2$.

Caution

A statistical test is not enough. Look for a cause (misread, bump, wrong start), *record* the rejection and report that you made it. Never discard a reading only because it is inconvenient.

Rule: independent contributions combine in quadrature

If a result depends on several *independent* sources, with standard uncertainties $u_1, u_2, \dots$ in the same unit, the combined standard uncertainty is

$$u_c = \sqrt{u_1^2 + u_2^2 + \dots}$$

Justification on the next slides (and in the supplementary note).

Physical meaning

The largest term dominates: $\sqrt{0.0232^2 + 0.0058^2} = 0.0239$ barely differs from 0.0232. **It is pointless to improve a small source while a large one remains.** This is the single most useful habit in laboratory work.

An opening question: how does δL and δT become δg ?

Pendulum formula

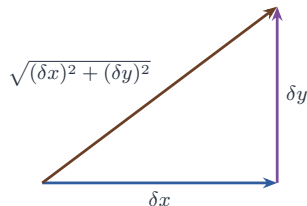
$$g = \frac{4\pi^2 L}{T^2}, \quad L = 0.800 \pm 0.002 \text{ m}, \quad T = 1.795 \pm 0.005 \text{ s}.$$

Predict first

Percentage uncertainty of L ? Of T ? Because T is *squared*, does its uncertainty count once, twice or half? Which quantity do you expect to dominate δg ? Is $\delta g/g$ the sum of the percentages, or smaller?

We return to this after the rules: the answer is a worked example.

Why quadrature?



independent errors rarely both reach their worst value together, so the combined size is smaller

→ $\delta x + \delta y$: both at their worst

Two independent sources δx and δy would give the maximum combined uncertainty $\delta x + \delta y$ only if both were at their worst at once, in the same direction.

Independent errors are sometimes positive, sometimes negative, so they partly cancel. Their combined size behaves like the hypotenuse of a right triangle. The simple sum is an *upper bound*.

Rule: propagation for common operations (independent uncertainties)

Sum or difference $q = x \pm y$ $\delta q = \sqrt{(\delta x)^2 + (\delta y)^2}$ (*absolute* uncertainties)

Product or quotient $q = xy$
or x/y $\frac{\delta q}{|q|} = \sqrt{\left(\frac{\delta x}{|x|}\right)^2 + \left(\frac{\delta y}{|y|}\right)^2}$ (*relative*)

Power $q = x^n$ $\frac{\delta q}{|q|} = |n| \frac{\delta x}{|x|}$

Constant factor $q = Bx$ $\delta q = |B| \delta x$ (exact B ; for example $\frac{1}{2}$, $4\pi^2$, 2)

Caution

The same quantity appearing twice is *not* independent: treat $x \cdot x$ as the power x^2 , giving $2 \delta x/x$, not $\sqrt{2} \delta x/x$. A difference of two nearly equal numbers has a large *relative* uncertainty.

Rule: propagation for any function $q = f(x, y, \dots)$ (independent uncertainties)

$$\delta q = \sqrt{\left(\frac{\partial f}{\partial x} \delta x\right)^2 + \left(\frac{\partial f}{\partial y} \delta y\right)^2 + \dots}$$

Physical meaning

The term $(\partial f / \partial x) \delta x$ is the change in q produced by a change δx in x alone, with every other variable held fixed: the *contribution* of x . All four special cases are this formula applied to $x \pm y$, xy , x/y and x^n .

Partial derivative: differentiate with respect to one variable, treating the others as constants. For $f = L \sin \theta$: $\partial f / \partial L = \sin \theta$ and $\partial f / \partial \theta = L \cos \theta$. Angles must be in radians.

TOOLKIT 4 – Propagating uncertainty

1. **Write the formula** for the result. Check that the inputs are independent.
2. **Choose the rule:** sums and differences (absolute), products and powers (relative), anything else (partial derivatives).
3. **Compute each contribution separately** and tabulate them.
4. **Combine in quadrature.**
5. **Find the dominant contribution** (the largest). A contribution smaller than about a third of it hardly matters.
6. **Round** by Toolkit 1 conventions: one significant figure on δq , then the value.

CHECK: δq must be at least as large as the largest single contribution and at most their plain sum.

Lecturer example: g from the pendulum (relative-uncertainty rule)

IDENTIFY. $L = 0.800 \pm 0.002$ m,
 $T = 1.795 \pm 0.005$ s. Required: $g = 4\pi^2 L/T^2$
and δg .

SET UP. A product, a quotient and a power:
use relative uncertainties. *Prediction:*
 $g \approx 9.8$ m/s²; the squared T counts double.

EXECUTE.

$$g = \frac{4\pi^2(0.800)}{1.795^2} = 9.80 \text{ m/s}^2,$$

$$\frac{\delta L}{L} = \frac{0.002}{0.800} = 0.25\%,$$

$$2\frac{\delta T}{T} = 2 \times \frac{0.005}{1.795} = 0.56\%.$$

$$\frac{\delta g}{g} = \sqrt{0.25^2 + 0.56^2}\% = 0.61\%,$$

$$\delta g = 9.80 \times 0.0061 = 0.06 \text{ m/s}^2.$$

EVALUATE. $g = (9.80 \pm 0.06) \text{ m/s}^2$.

- T dominates: its 0.28% doubles to 0.56%.
- Check: 0.61% lies between 0.56% (largest) and 0.81% (plain sum). ✓
- Value agrees with 9.78: $D = +0.02 < 0.06$.
- The tool to improve: time more oscillations.

(a) Temperature rise. $T_i = 21.3 \pm 0.2^\circ\text{C}$,
 $T_f = 34.6 \pm 0.2^\circ\text{C}$.

Difference: absolute uncertainties in
quadrature:

$$\Delta T = 13.3^\circ\text{C},$$

$$\delta\Delta T = \sqrt{0.2^2 + 0.2^2} = 0.28^\circ\text{C}.$$

$\Delta T = (13.3 \pm 0.3)^\circ\text{C}$ (the plain sum 0.4 is
the upper bound). The relative uncertainty
has grown: $0.2/34.6 = 0.6\%$ became
 $0.28/13.3 = 2.1\%$.

(b) A general formula. A ramp has length
 $L = 1.200 \pm 0.002\text{ m}$ at angle $\theta = 30.0^\circ \pm 0.5^\circ$.
Height $h = L \sin \theta$.

$h = 1.200 \sin 30.0^\circ = 0.600\text{ m}$. With
 $\delta\theta = 0.5^\circ = 0.00873\text{ rad}$:

$$\sin \theta \delta L = 0.0010,$$

$$L \cos \theta \delta\theta = 0.0091,$$

$$\begin{aligned}\delta h &= \sqrt{0.0010^2 + 0.0091^2} \\ &= 0.009\text{ m}.\end{aligned}$$

$h = (0.600 \pm 0.009)\text{ m}$; the angle dominates.

Select the rule, predict which measurement dominates, then calculate

A brass cylinder has mass $m = 95.2 \pm 0.1$ g, diameter $d = 1.250 \pm 0.005$ cm and height $h = 8.45 \pm 0.02$ cm. The density is $\rho = \frac{m}{V} = \frac{4m}{\pi d^2 h}$.

1. Calculate ρ and $\delta\rho$ and write the result in full.
2. Which of the three measurements should be improved first? By how much would $\delta\rho$ fall if its relative uncertainty were halved?

Rule: product, quotient, power: relative uncertainties. *Prediction:* d enters squared and has the largest relative uncertainty (0.4%), so it dominates.

$$\rho = \frac{4(95.2)}{\pi(1.250)^2(8.45)} = 9.18 \text{ g/cm}^3.$$

Source	Relative contribution	Value
m	$0.1/95.2$	0.105%
d (squared)	$2 \times 0.005/1.250$	0.80%
h	$0.02/8.45$	0.24%

$$\frac{\delta\rho}{\rho} = \sqrt{0.105^2 + 0.80^2 + 0.24^2}\% = 0.84\%, \quad \delta\rho = 0.077 \rightarrow 0.08 \text{ g/cm}^3.$$

$\rho = (9.18 \pm 0.08) \text{ g/cm}^3$. The diameter dominates (about 90% of the variance). Halving $\delta d/d$ gives $\sqrt{0.105^2 + 0.40^2 + 0.24^2} = 0.48\%$, so $\delta\rho \approx 0.04 \text{ g/cm}^3$.

First-level rules (OpenStax §1.6)

- Multiplying: keep the fewest significant figures.
- Adding: keep the fewest decimal places.
- Quick, needs no uncertainties, but *hides* what dominates and can mislead.

Propagation

- Uses the actual δx of each quantity.
- Reveals the dominant source.
- Is the standard for any reported laboratory result.

Example where the shortcut fails

$\rho = 9.18 \text{ g/cm}^3$ has three significant figures from the density example. Its real uncertainty is 0.08, so the second decimal is already doubtful: the sig-fig rule alone could not say this. A difference $34.6 - 21.3 = 13.3$ also shows ΔT with three digits while its relative uncertainty is 2%.

Working habit: carry two or three extra digits through the calculation and round once, at the end, using the uncertainty.

Can you now:

- calculate $\bar{x}$, s and $s/\sqrt{N}$, and say what each means?
- decide about an outlier, and report the decision?
- combine independent uncertainties in quadrature?
- propagate through sums, products, powers and a general formula?
- find the dominant source and say how to improve the result?

Next

So far each result came from one formula. When we vary a quantity deliberately and fit a graph, the uncertainty of the *slope* is what we need. Then everything goes into one budget.

Contact 3: Apply

Graphs, budgets, comparison with theory and the report

An opening question: a graph in place of one number

Hooke's law: hang masses on a spring, measure the length l with a ruler (limit 0.1 cm on each reading)

m (g)	50	100	150	200	250	300	350	400
l (cm)	14.1	16.0	17.9	20.1	21.8	23.8	25.9	27.6

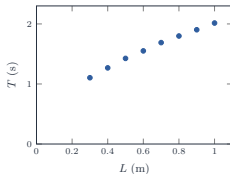
Predict first

Sketch l against m . Is it a straight line through the origin? What do the slope and the intercept mean physically? Would a single measurement of k from one mass be as trustworthy as the fit?

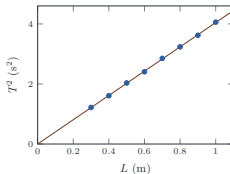
The model: $l = l_0 + (g/k) m$, with l_0 the unstretched length and k the spring constant. The data are illustrative.

Toolkit 5: linearising and fitting

(a) T against L : a curve



(b) T^2 against L : a straight line



TOOLKIT 5 – Linearise, then fit

1. Write the model; rearrange to $y = mx + c$.
2. Plot y against x with error bars.
3. Fit; read slope, intercept and their uncertainties.
4. Convert the slope to the physical quantity (propagate).

Relation	Plot	Slope
$T^2 = (4\pi^2/g) L$	T^2 against L	$4\pi^2/g$
$l = l_0 + (g/k) m$	l against m	g/k
$P = nRT (1/V)$	P against $1/V$	nRT

- An **error bar** of length $\pm\sigma_y$ on each point shows the uncertainty of that reading.
- About two thirds of the bars should touch the line: a line missing all bars, or touching all with room to spare, signals trouble.
- **By hand:** draw the best line, then the steepest and shallowest lines consistent with the bars. The slope range gives δm . Good for checking.
- **By calculation:** least squares (next slide). It chooses the line that minimises the sum of squared vertical distances.

Why a fit helps

Each point carries a small uncertainty. The line uses *all* the points, so its slope is better determined than any single ratio l/m . It also reveals an offset (the intercept) that a single reading would hide.

Rule: least-squares line $y = mx + c$, equal uncertainty σ_y on every y

$$\Delta = N \sum x^2 - \left(\sum x \right)^2, \quad m = \frac{N \sum xy - \sum x \sum y}{\Delta}, \quad c = \frac{\sum x^2 \sum y - \sum x \sum xy}{\Delta}.$$

$$\delta m = \sigma_y \sqrt{\frac{N}{\Delta}}, \quad \delta c = \sigma_y \sqrt{\frac{\sum x^2}{\Delta}}.$$

If σ_y is unknown, estimate it from the scatter about the line:

$$\sigma_y \approx s_y = \sqrt{\frac{\sum (y_i - mx_i - c)^2}{N - 2}}.$$

Assumptions: the uncertainty is in y only (the x values are much better known), is the same for every point, and the points are independent. Weighted fits, when the σ_i differ, are in the supplementary note.

Worked example: the spring

IDENTIFY. $N = 8$ points (m, l) ; $\sigma_l = 0.1$ cm.

Required: slope, intercept, k .

SET UP. Line $l = c + (\text{slope}) m$. *Prediction:*
slope near 0.04 cm/g (about 2 cm per 50 g);
intercept near 12 cm.

EXECUTE. $\sum m = 1800$ g,

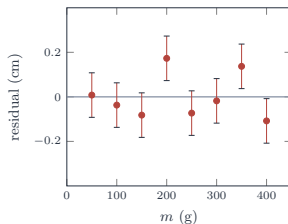
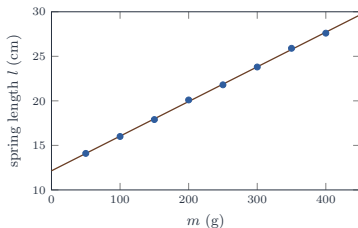
$$\sum m^2 = 5.10 \times 10^5 \text{ g}^2,$$

$$\Delta = 8(5.10 \times 10^5) - 1800^2 = 8.4 \times 10^5 \text{ g}^2.$$

The formulas give

$$\text{slope} = (0.0389 \pm 0.0003) \text{ cm/g},$$

$$c = (12.15 \pm 0.08) \text{ cm}.$$



EVALUATE. Predictions agree. The line passes near all bars (left). The intercept is the unstretched length, 12.15 ± 0.08 cm. The residuals (right) scatter about zero with no pattern.

Rule: reduced chi-squared

$$\chi^2 = \sum_i \left(\frac{y_i - mx_i - c}{\sigma_i} \right)^2, \quad \nu = N - 2, \quad \chi_{\text{red}}^2 = \frac{\chi^2}{\nu}.$$

Rule of thumb: $\chi_{\text{red}}^2 \approx 1$: the model fits within the stated uncertainties. $\chi_{\text{red}}^2 \gg 1$: the model is wrong, or the σ_i are too small. $\chi_{\text{red}}^2 \ll 1$: the σ_i are too large, or the data have been fitted too well.

Spring: $\chi^2 = 7.4$, $\nu = 6$, $\chi_{\text{red}}^2 = 1.23$. Acceptable (the probability of exceeding this value by chance is 0.29).

Residual plot: always draw it. A random scatter about zero supports the model; a curve or a trend means the model is wrong, even if χ_{red}^2 looks fine.

IDENTIFY. Slope = g/k with $g = 9.78 \text{ m/s}^2$ (uncertainty negligible). Required: k .

SET UP. Convert the slope to SI:

$1 \text{ cm/g} = 10 \text{ m/kg}$, so

$0.0389 \text{ cm/g} = 0.389 \text{ m/kg}$. $k = g/\text{slope}$ is a quotient with an exact numerator, so $\delta k/k = \delta(\text{slope})/\text{slope} = 0.0003/0.0389 = 0.8\%$.

EXECUTE.

$$k = \frac{9.78}{0.389} = 25.1 \text{ N/m},$$

$$\delta k = 0.008 \times 25.1 = 0.2 \text{ N/m}.$$

EVALUATE. $k = (25.1 \pm 0.2) \text{ N/m}$. Units: $(\text{m/s}^2)/(\text{m/kg}) = \text{kg/s}^2 = \text{N/m}$. ✓

Compare two methods

A second experiment, timing the oscillations, gives $k = 24.6 \pm 0.3 \text{ N/m}$. The difference is 0.5; the combined uncertainty is $\sqrt{0.2^2 + 0.3^2} = 0.36$, so the difference is 1.4 combined uncertainties: the two methods agree (rule on a later slide).

TOOLKIT 6 – Building an uncertainty budget

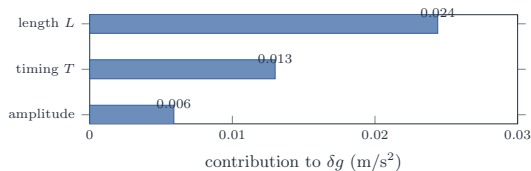
A table with one row per source, filled *before* any final claim:

1. **Source**, with its Type (A or B) and whether random or systematic.
2. **Standard uncertainty** of the input, from Toolkit 2.
3. **Contribution to the result**: $|\partial f / \partial x| \delta x$ (or the relative rule).
4. **Share of the total variance**: $(\text{contribution})^2 / (\delta q)^2$.
5. **Total** in quadrature, the **dominant source**, and what to do about it.

CHECK: systematic effects are *corrected* first, then their small remaining uncertainty enters as a row.

The whole experiment: g from the pendulum

Source	Type	Input	Contribution to g	Variance share
Length $L = 0.800$ m	B	0.002 m	0.024	75%
Period $T = 1.7997$ s	A, B	0.0012 s	0.013	21%
Amplitude correction ($8^\circ \pm 1^\circ$)	B	$\pm 1^\circ$	0.006	4%
Total (quadrature)			0.028	100%



Reading the budget. The *length* dominates. The uncertainty of T from ten timings is already small; further repeats would help little. To improve the result, measure L better: a longer rule, or a better way to locate the bob centre.

A systematic correction: finite amplitude

IDENTIFY. The formula $g = 4\pi^2 L/T^2$ assumes small swings. The measured period at amplitude θ_0 is longer: $T = T_0 (1 + \theta_0^2/16)$ (first correction, θ_0 in radians). So g from the raw T is too small.

SET UP. $\theta_0 = 8^\circ = 0.1396$ rad. *Prediction:* the correction is a fraction of one percent, raising g slightly.

EXECUTE. Raw:

$$g_{\text{raw}} = 4\pi^2(0.800)/1.7997^2 = 9.751 \text{ m/s}^2.$$

$$\text{Correction factor } (1 + \theta_0^2/16)^2 = 1.00244:$$

$$g = 9.775 \text{ m/s}^2. \text{ Uncertainty of the correction (from } \pm 1^\circ \text{ in } \theta_0): 0.006.$$

EVALUATE. Combined with the other two rows:

$$\delta g = \sqrt{0.024^2 + 0.013^2 + 0.006^2} = 0.028.$$

$$g = (9.77 \pm 0.03) \text{ m/s}^2$$

- The correction (+0.024) is about as large as the total uncertainty, so it *must* be applied and reported.
- Without it the raw 9.75 ± 0.03 would still agree with 9.78, but only because the budget is generous.

Rule: comparing two results (or a result and an accepted value)

For results $a \pm \delta a$ and $b \pm \delta b$ define

$$n = \frac{|a - b|}{\sqrt{(\delta a)^2 + (\delta b)^2}} \quad (\delta b = 0 \text{ if } b \text{ is an accepted value}).$$

$n \lesssim 2$: **consistent**. $2 \lesssim n \lesssim 3$: **doubtful**; look harder. $n \gtrsim 3$: **inconsistent**; a cause is likely.

Comparison	n	Verdict
Pendulum: 9.77 ± 0.03 against accepted 9.78	$0.01/0.03 = 0.3$	consistent
Spring: static 25.1 ± 0.2 against dynamic 24.6 ± 0.3	$0.5/0.36 = 1.4$	consistent
Student B: 9.62 ± 0.04 against accepted 9.78	$0.16/0.04 = 4$	inconsistent

Caution. Agreement does not prove a theory right; disagreement does not prove it wrong. Look first for a missing source, an underestimated uncertainty or a false assumption.

Plan a better pendulum measurement

The target is $\delta g \leq 0.015 \text{ m/s}^2$ (half of today's 0.028). Using the budget, what should be improved, and by how much?

Reasoning. L contributes 0.024: halving δL to 0.001 m gives 0.012. T contributes 0.013: 40 timings in place of 10 halve it to 0.006. The amplitude term (0.006) stays.

$$\begin{aligned}\delta g &= \sqrt{0.012^2 + 0.006^2 + 0.006^2} \\ &= 0.015 \text{ m/s}^2.\end{aligned}$$

Lesson

- Improving T alone leaves $\sqrt{0.024^2 + 0.006^2 + 0.006^2} = 0.025$: almost nothing gained.
- Improving L alone gives $\sqrt{0.012^2 + 0.013^2 + 0.006^2} = 0.019$: not enough.
- Both are needed. The budget tells you *before* the experiment is repeated.

Specific heat capacity of a metal block

A metal block of mass 0.150 ± 0.001 kg is heated to 98.0 ± 0.5 °C and dropped into 0.200 ± 0.001 kg of water at 21.3 ± 0.2 °C in an insulated container. The final temperature is 30.5 ± 0.2 °C. Take $c_{\text{water}} = 4186$ J/(kg K) (exact) and neglect the container.

1. Predict whether the block's temperature fall is larger or smaller than the water's temperature rise, and why.
2. Find c for the metal and its uncertainty. Which measurement dominates?
3. The accepted value for aluminium is 900 J/(kg K). Is the block aluminium? State any systematic effect you suspect.

Use IDENTIFY, SET UP, EXECUTE, EVALUATE. Choose the rules yourself.

IDENTIFY and SET UP. Heat lost by the block equals heat gained by the water, $c m_s (T_s - T_f) = m_w c_w (T_f - T_w)$, so

$$c = \frac{m_w c_w (T_f - T_w)}{m_s (T_s - T_f)}.$$

Prediction: the block's mass is smaller and c is smaller than the water's, so its temperature change is larger: about 67.5°C against 9.2°C .

EXECUTE. $c = \frac{0.200(4186)(9.2)}{0.150(67.5)} = 761 \text{ J/(kg K)}$. Contributions (each variable in turn, others fixed): m_w : 4; T_w : 17; m_s : 5; T_s : 6; T_f : 19 (all in J/(kg K)). Total in quadrature = 26.

EVALUATE. $c = (760 \pm 30) \text{ J/(kg K)}$. The final temperature (0.2 K on a 9.2 K rise) and the water temperature dominate. Against 900: $n = 139/26 \approx 5$: **inconsistent**. A systematic cause: heat lost to the container and to the air makes the water seem to gain less heat, giving a low c . The block may be aluminium, but a better experiment (insulate, include the calorimeter's heat capacity, larger temperature rise) is needed to say.

Reporting the result in the laboratory report

A usual structure (confirm the format your course requires):

1. Title, aim, theory (formula and assumptions).
2. Apparatus, with each instrument's resolution.
3. Method, including steps taken to reduce uncertainty.
4. **Data** in tables: quantity, unit and uncertainty in each column heading.
5. **Analysis**: sample calculation, propagation, graph with error bars, fit parameters.
6. **Result** in full reporting form with percent uncertainty.
7. **Discussion**: agreement (n), budget, dominant source, systematic effects, improvements.
8. Conclusion.

✓ Checklist before handing in

- Every number has a unit.
- Every result has an uncertainty, rounded correctly.
- Systematic effects were looked for and stated.
- Graph: labelled axes with units, error bars, fitted line, residuals.
- The budget names the dominant source.
- The comparison states n , not just “close”.
- The conclusion says what was learned, not only the number.

In the data

- Writing 3.470 ± 0.05 (digits finer than the uncertainty).
- Forgetting the zero error.
- Timing one oscillation instead of 20.
- Averaging a drifting quantity.

In the analysis

- Adding uncertainties linearly instead of in quadrature.
- Forgetting the exponent: T^2 counts double.
- Rounding intermediate steps.
- Reporting s where $s/\sqrt{N}$ is meant (or the reverse).
- Using degrees where radians are needed.
- Blaming “human error” without a quantitative estimate.

Backwards: OpenStax §1.6 introduced accuracy, precision, discrepancy and percent uncertainty. We kept those terms and added the standard laboratory practice: standard uncertainties, quadrature, least squares and the budget.

Forwards: every experiment in PHYS 105 (momentum, motion, force, circular and rotational motion, gravitation, heat) ends with a result, an uncertainty and a comparison. The same toolkit serves in all of them.



Laboratory habits from this lecture

- Write the uncertainty beside each reading, at the bench.
- Look for systematics before you repeat.
- Find the dominant source early and spend your effort there.
- Round last. Compare in terms of n .

I can ...

1. report a result as (value $\pm$ uncertainty) unit with correct rounding and percent uncertainty;
2. distinguish accuracy from precision, random from systematic, Type A from Type B;
3. calculate $\bar{x}$, s and $s/\sqrt{N}$, and judge an outlier;
4. propagate uncertainty through sums, products, powers and general formulas;
5. fit a straight line and interpret slope, intercept, residuals and χ^2_{red} ;
6. build a budget, test agreement with n , and write the report.

Toolkits this lecture

- T1 Reading an instrument
- T2 Classifying a source
- T3 Summarising repeated data
- T4 Propagating uncertainty
- T5 Linearising and fitting
- T6 Building a budget and comparing results

Next: the tutorial sheet (answers at the end) and the supplementary note (derivations, tables, weighted fits).